STAFF REVIEW · H446 064 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A task requires another task's result.

2. Amount of completed work per unit time.

3. Result depends on unsafe timing/order of shared operations.

4. Only a rising edge samples D. Changing D and reaching a falling edge do not update Q, so it holds 1 until the next rising edge samples 0.

I Do — source worked steps

1. Explain the decision — Only independent tasks can safely overlap without waiting for each other.

We Do — guided CHECK

1. Explain the decision
Extra processors do not remove logical dependency.

You Do — source mark schemes

1. Sequential 11s [2]; parallel max7,4=7s [2].

2. Independent image load and help-page download [2]; validate/authorise then commit payment/order sequential [2], justify dependencies [2].

3. Both read same old value [2], both write overlapping update [1], one increment lost [1], lock/atomic increment [2].

Exit Ticket — source answer

1. False. Dependencies and overhead constrain speedup.